Type Traits and Compile-time C++

Ask questions about types and compute values while compiling: `<type_traits>`, `enable_if`, `if constexpr`, `constexpr` and template metaprogramming.

Type traits

A **type trait** answers a question about a type at compile time (`::value` / `_v`) or transforms one (`::type` / `_t`).

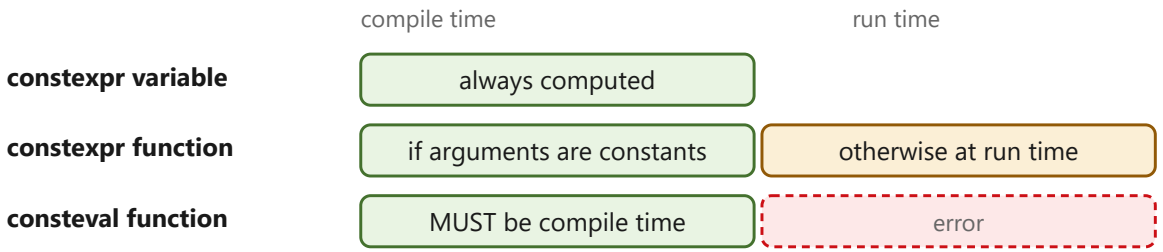
```
#include <type_traits>

static_assert(std::is_integral_v<int>);                        // query: true
static_assert(!std::is_integral_v<double>);
static_assert(std::is_same_v<int, std::remove_reference_t<int&>>);   // transform, then compare

template <typename T>
T multiply(T a, T b) {
    static_assert(std::is_arithmetic_v<T>, "multiply needs a number type");
    return a * b;
}
// multiply(std::string("a"), std::string("b"));   // compile error with YOUR message
```

KEY POINT `static_assert`: compile-time, free. `assert`: run-time, gone when `NDEBUG` is defined.

`constexpr`, `consteval`, `constinit`



constinit: initialised at compile time, still mutable (avoids static-init-order problems).

constexpr does not force compile time; assign to a constexpr variable or use consteval.

Write an ordinary function and ask for compile-time evaluation.

KEYWORD	MEANING
`constexpr` variable	Must be initialised with a constant expression; usable where constants are required (array sizes, template arguments).
`constexpr` function	*Can* run at compile time if its arguments are constants; can still be called at run time with ordinary values.
`consteval` (C++20)	An "immediate" function: every call **must** be evaluated at compile time.
`constinit` (C++20)	The variable is initialised at compile time (no static-initialisation-order problems), but remains mutable.

```
constexpr int factorial(int n) { return n ≤ 1 ? 1 : n * factorial(n - 1); }

constexpr int f10 = factorial(10);        // computed by the compiler
int arr[factorial(4)];                    // array of 24: needs a constant
expression

int n = 5;
int r = factorial(n);                     // fine: runs at run time, because
n is not a constant

consteval int sq(int x) { return x * x; }
constexpr int a = sq(7);                  // OK
// int b = sq(n);                         // ERROR: n is not a constant
expression
```

`enable_if` and SFINAE

SFINAE: an invalid substitution drops that overload instead of erroring. `std::enable_if` uses it to switch overloads by a condition.

```
template <typename T, std::enable_if_t<std::is_integral_v<T>, bool> = true>
void print(const T& v) { std::cout << "integral: " << v << '\n'; }

template <typename T, std::enable_if_t<!std::is_integral_v<T>, bool> = true>
void print(const T& v) { std::cout << "other: " << v << '\n'; }

print(42);      // first overload
print(2.5);     // second overload: for double the first overload's enable_if
has no ::type, so it is discarded
```

Noisy syntax, poor errors. Better tools:

TOOL	STANDARD	LOOKS LIKE
`if constexpr`	C++17	Branch inside *one* function; the untaken branch is not even compiled for that type
Concepts / `requires`	C++20	`template <std::integral T> void print(T v);`

```cpp
template <typename T>
void print(const T& v) {
    if constexpr (std::is_integral_v<T>) std::cout << "integral: " << v <<
'\n';
    else                                 std::cout << "other: " << v << '\n';
}

template <std::integral T>        // C++20: constrained template, readable
errors
T twice(T v) { return v * 2; }
```

> **PLAIN IF IS NOT ENOUGH** With `if (is_integral_v<T>)` **both** branches must compile for every `T`. `if constexpr` discards the false one.

Concepts and `requires` (C++20)

A **concept** is a named, compile-time condition on a type. It replaces `enable_if` with something you can read, and it turns a page of substitution errors into one clear message at the call.

```cpp
template <typename T>
concept Addable = requires(T a, T b) { { a + b } -> std::convertible_to<T>;
};   // what must compile, and what it must give

template <Addable T>                    // 1. constrained template parameter
T sum2(T a, T b) { return a + b; }

template <typename T>
requires std::integral<T>               // 2. a requires clause
T twice(T v) { return v * 2; }

void show(std::integral auto x)       { std::cout << "integral " << x <<
'\n'; }       // 3. constrained auto (abbreviated)
void show(std::floating_point auto x) { std::cout << "floating " << x <<
'\n'; }      //    overloads by concept

static_assert(Addable<int> && Addable<std::string> && !Addable<NoPlus>);
// concepts are compile-time booleans
sum2(2, 3);                              // 5
sum2(std::string("a"), std::string("b"));   // "ab"
```

```
twice(21);                                  // 42 (compiled)
show(7);  show(2.5);                        // integral 7 / floating 2.5
// twice(2.5);   // error: no matching function, constraint
'std::integral<double>' not satisfied
```

- A **requires-expression** (`requires(T a) { ... }`) lists operations that must compile. `{ expr } -> Concept` also checks the result type.
- The standard library ships many: `std::integral`, `std::floating_point`, `std::same_as`, `std::convertible_to`, `std::invocable`, `std::regular`, and the range concepts such as `std::ranges::range`.
- **Overloads** can differ only by concept; when two are satisfied, the more constrained one wins.
- A concept checks *syntax and types*, not behaviour: `Addable` does not promise that `+` is commutative. Document the meaning you expect.

TOOL	ERROR WHEN IT FAILS	USE FOR
`enable_if` / SFINAE	pages of "no matching function" notes	pre-C++20 code
`static_assert` on a trait	your message, but inside the function body	a single check with a custom message
concept	"constraint not satisfied" at the call site	the interface of a template (C++20 and later)

Template metaprogramming (TMP)

Before `constexpr` loops, compile-time computation was **recursive template instantiation**. You will meet it in older code.

```
template <int N>
struct Fact {
    static constexpr int value = Fact<N - 1>::value * N;    // the compiler
instantiates Fact<9>, Fact<8>, ...
};
template <>
struct Fact<0> {                                             // full
specialisation ends the recursion
    static constexpr int value = 1;
};

static_assert(Fact<5>::value == 120);
```

Each `Fact<N>` is a separate class; the result is a baked-in constant.

- **Limits:** recursion depth cap (a few hundred), slow builds, long errors.
- **Today:** `constexpr` for values, traits + `if constexpr`/concepts for types; TMP only to compute *types*.

TASK	OLD STYLE	MODERN STYLE
Compute a value	recursive struct with `::value`	`constexpr` function
Pick code by type	`enable_if` overloads	`if constexpr` or concepts
Reject bad types	obscure substitution errors	`static_assert` / `requires`

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
Type traits	Programming: Principles and Practice	18.1.3 Concepts	531 to 533
	Programming: Principles and Practice	19.4 and 21.1 Generalizing Vector again; standard algorithms	573 to 578 and 617 to 622
constexpr, consteval, constinit	Effective Modern C++	Item 15, Use constexpr whenever possible	115 to 121
enable_if and SFINAE	Effective Modern C++	Items 26 to 27, Overloading on universal references and its alternatives (tag dispatch, enable_if)	195 to 215